

FARMCAST

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List of Abbreviations

Abbreviation	Meaning
ML	Machine Learning
ICT	Information & Communication Technology
CLR	Cow Lactometer Reading
SNF	Solid-Not-Fat
UI	User Interface
MVC	Model-View-Controller
CSV	Comma-Separated Values
DB	Database
API	Application Programming Interface
MILMA	Kerala Co-operative Milk Marketing Federation
RF	Random Forest
LR	Logistic Regression

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Abstract

FarmCast is an integrated, intelligent dairy decision-support system that combines machine learning, web technologies, and domain-specific datasets to enhance the efficiency of dairy farm management. The system is developed as a Django-based web application that allows farmers to input real-time milk analyzer parameters and instantly receive insights related to milk quality, pricing, cattle breed suitability, and feed recommendations. Unlike traditional approaches that rely on manual judgement, FarmCast automates the full analytical workflow from data preprocessing and model prediction to user-friendly result presentation.

The system uses live milk analyzer readings such as CLR, fat percentage, SNF, and temperature to classify milk into quality grades using trained classification models. A regression model predicts fair milk prices by incorporating seasonal patterns and historical data. The breed prediction module identifies the most suitable cattle breed for a given region based on yield characteristics and environmental tolerance. Additionally, the system provides feed recommendations dynamically linked to the predicted breed, improving nutritional planning and milk productivity. These ML models are integrated through Django views, pipelines, and pre-trained joblib files, enabling seamless execution from the web interface.

FarmCast employs a structured architecture that includes data preprocessing, feature engineering, model training, and real-time inference through the web interface. The system provides clear visual outputs, structured prediction pages, and automated transitions between forms and result screens. By merging data analytics, agriculture expertise, and modern web development, FarmCast offers a practical and scalable solution for dairy farmers. It supports informed decision-making, reduces dependency on manual assessments, and improves both productivity and profitability. Future extensions include IoT-based automated milk testing, multilingual mobile app integration, and real-time weather-informed analytics.

1. Problem Definition

1.1 Overview

Dairy farmers often struggle with accurately assessing milk quality, determining fair market prices, selecting cattle breeds suitable for their regional climate, and choosing the right feed to maximize milk yield. Traditional decision-making relies heavily on manual judgment, which can lead to inconsistent results, reduced productivity, and financial losses. With the availability of digital data from milk analyzers, breed profiles, and feed information, there is a strong need for an intelligent system that can process this data and generate reliable recommendations.

FarmCast addresses this challenge by integrating machine learning models into a single, user-friendly platform that automates prediction and analysis. The system evaluates milk parameters to classify quality, predicts fair prices using historical patterns, recommends appropriate cow breeds based on environmental suitability, and provides feed suggestions tailored to the predicted breed. By converting raw farm data into actionable insights, FarmCast supports informed decision-making and improves overall dairy farm efficiency.

1.2 Problem Statement

Dairy farmers face significant challenges in accurately evaluating milk quality, determining fair prices, selecting cattle breeds suitable for local environmental conditions, and choosing appropriate feed to optimize milk production. These decisions are often made based on subjective judgment, limited experience, or inconsistent traditional methods, resulting in inaccurate pricing, low productivity, poor breed performance, and inefficient feeding practices. Although large amounts of dairy-related data are increasingly available from milk analyzers, breed characteristics, and feed composition farmers lack an integrated system that can automatically analyze this data and provide reliable, data-driven recommendations.

There is a need for an intelligent, automated platform that combines machine learning, agricultural knowledge, and real-time data processing to support farmers in making informed decisions across multiple aspects of dairy management. The problem is to design and develop a system that can classify milk quality, predict fair milk prices, recommend optimal cattle breeds, and suggest suitable feed options—all within a single, user-friendly solution.

2. Introduction

The dairy industry is one of the most crucial sectors of agriculture, supporting millions of farmers who rely on milk production as a primary source of income. Despite its significance, many critical decisions in dairy farming—such as assessing milk quality, determining fair prices, selecting suitable cattle breeds, and planning nutritional feed—are still made through subjective judgment and inconsistent traditional practices. These manual approaches often result in financial losses, reduced productivity, poor breed performance, and inefficient feed utilization. With the growing availability of digital tools and data-driven technologies, there is a strong need for intelligent systems capable of transforming raw dairy data into reliable and actionable insights.

FarmCast is developed to meet this need. It is a Django-based intelligent dairy data analysis system that integrates machine learning models into a user-friendly web platform. The system processes real-time milk analyzer parameters such as CLR, SNF, fat content, and temperature to classify milk into standardized quality grades. It predicts fair market prices by incorporating historical pricing patterns and seasonal variations. The breed prediction module evaluates environmental suitability, yield characteristics, and regional climate data to recommend the most appropriate cattle breed for a given region. A key advancement of FarmCast is its integrated feed recommendation module, which automatically generates breed-specific feed suggestions immediately after the breed prediction step, helping farmers optimize nutritional planning and enhance milk productivity.

The system architecture combines data collection, preprocessing, model training, and real-time prediction using classification, regression, and recommendation algorithms. These models are embedded within Django views and delivered through an interactive and intuitive web interface. Farmers can input their data, receive instant predictions, and make informed decisions without requiring any technical expertise. By merging machine learning, dairy science, and modern web technologies, FarmCast provides a comprehensive and practical solution that enhances operational efficiency, reduces uncertainty, and supports sustainable dairy management.

As dairy farming progresses toward digitization and precision agriculture, FarmCast serves as a scalable platform with potential for future enhancements such as IoT-enabled milk testing, multilingual mobile app integration, and weather-driven predictive analytics. The system represents a significant step toward smart dairy farming by empowering farmers with accurate predictions and intelligent recommendations.

3. Literature Survey

[1]. Sharma, R., & Singh, A. (2015). *Milk Quality Assessment Using Physico-Chemical Parameters*. Journal of Dairy Science Research.

The paper “Milk Quality Assessment Using Physico-Chemical Parameters” by Sharma and Singh (2015) focuses on evaluating milk quality using fundamental laboratory measurements. Since milk is highly perishable and easily adulterated, the authors analyze key physico-chemical parameters such as fat content, SNF, protein, lactose, pH, acidity, density, and electrical conductivity. Each of these parameters reflects important aspects of milk composition and safety. Fat, SNF, protein, and lactose indicate nutritional value, while pH and acidity reveal freshness and microbial activity. Density and conductivity help identify adulteration, especially the addition of water or salts. The study describes standard testing procedures and compares the results with established dairy quality standards. By identifying deviations from normal ranges, the authors demonstrate how physico-chemical profiling can effectively detect spoilage and low-quality or adulterated milk. The paper concludes that these parameters offer a reliable, simple, and cost-efficient method for routine milk quality monitoring in the dairy industry.

[2]. Patil, P., & Pawar, M. (2019). *Machine Learning Techniques for Milk Purity*

Classification. International Journal of Computer Applications.

The paper “Machine Learning Techniques for Milk Purity Classification” by Patil and Pawar (2019) explores how machine learning can be applied to automatically determine the purity of milk. Since milk adulteration is a major concern in the dairy supply chain, the authors focus on detecting impurities by analyzing measurable parameters such as fat content, SNF, pH, conductivity, and density. The study examines different machine learning algorithms—including decision trees, support vector machines (SVM), k-nearest neighbors (KNN), and artificial neural networks—to classify milk samples as pure or adulterated. The authors compare the accuracy of these models and highlight which technique provides the most reliable classification performance. Their results demonstrate that machine learning can significantly improve the speed and accuracy of milk quality evaluation compared to manual laboratory methods. The paper concludes that ML-based systems offer a promising and efficient approach for real-time milk purity monitoring in dairy industries.

[3]. Khan, S. et al. (2020). *SVM-Based Detection of Milk Adulteration*. IEEE Access.

The paper “SVM-Based Detection of Milk Adulteration” by Khan et al. (2020), published in *IEEE Access*, explores the use of machine learning, specifically Support Vector Machines (SVM), to detect adulteration in milk. Milk adulteration is a significant problem in the dairy industry, where substances such as water, starch, urea, or detergents are added to increase volume or simulate quality, posing health risks to consumers. The study aims to develop an automated, accurate, and efficient method for identifying adulterated milk samples. In the research, milk samples with different types and levels of adulterants were collected, and several key physico-chemical features were measured, including fat content, SNF (solids-not-fat), pH, conductivity, density, and freezing point. These features were then used to train an SVM classifier. The authors experimented with various kernel functions and hyperparameter settings to optimize the model’s performance. The results demonstrate that the SVM model can effectively classify milk samples as pure or adulterated with high accuracy, sensitivity, and specificity, outperforming traditional manual testing methods. The study concludes that SVM-based detection provides a reliable, rapid, and cost-effective solution for real-time monitoring of milk quality, offering practical applications for dairy industries and regulatory authorities to ensure consumer safety and product integrity.

[4]. Kumar, R., & Devi, S. (2018). *Statistical Analysis of Milk Price Trends in India*.

Agricultural Economics Review.

The paper “Statistical Analysis of Milk Price Trends in India” by Kumar and Devi (2018), published in *Agricultural Economics Review*, investigates the historical trends and patterns in milk prices across different regions of India. The study highlights the economic importance of the dairy sector, which is a significant source of livelihood for millions of farmers and a major contributor to the country’s agricultural GDP. Understanding milk price fluctuations is essential for policy-making, market planning, and ensuring fair income for dairy producers. The authors collect and analyze extensive data on milk prices from various states over multiple years. Using statistical techniques such as time series analysis, regression models, and descriptive statistics, they identify trends, seasonal variations, and regional disparities in milk prices. The study reveals factors influencing price changes, including production levels, demand fluctuations, transportation costs, and policy interventions. The findings show that milk prices are highly sensitive to both market dynamics and seasonal effects, with noticeable variation between urban and rural markets. The paper concludes that statistical monitoring of milk price trends can help policymakers, farmers, and dairy cooperatives make informed

decisions, improve price stability, and enhance the efficiency and profitability of the dairy supply chain in India.

[5]. Panda, D., & Sahu, R. (2021). *Random Forest Approach for Milk Price*

Prediction. Journal of Data Science and Analytics.

The paper “Random Forest Approach for Milk Price Prediction” by Panda and Sahu (2021), published in *Journal of Data Science and Analytics*, presents a machine learning method to predict milk prices accurately. Milk prices fluctuate due to factors like seasonal changes, production levels, demand, transportation costs, and market dynamics, making prediction important for farmers, traders, and policymakers. The study collects historical milk price data along with relevant factors such as production volume, demand trends, regional markets, and climatic conditions. The authors use a Random Forest regression model, an ensemble learning technique that constructs multiple decision trees and combines their outputs to enhance prediction accuracy and reduce overfitting. The model is trained and tested on the dataset, showing strong performance in forecasting milk prices under varying market conditions. The study concludes that Random Forest provides a reliable, efficient, and practical solution for milk price prediction, supporting informed decision-making, risk management, and better planning in the dairy industry.

[6]. Bharadwaj, K. (2017). *Breed Selection in Indian Dairy Farms: A Comparative*

Study. Indian Journal of Animal Sciences.

The paper “Breed Selection in Indian Dairy Farms: A Comparative Study” by Bharadwaj (2017), published in the *Indian Journal of Animal Sciences*, examines the factors influencing breed selection in Indian dairy farms. Choosing the right breed is critical for maximizing milk production, ensuring disease resistance, and optimizing feed efficiency. The study compares commonly reared dairy breeds in India, analyzing characteristics such as average milk yield, fat content, SNF (solids-not-fat), climate adaptability, disease resistance, and feed requirements. The author gathers data from various dairy farms across different regions to evaluate breed performance under local environmental and management conditions. The study highlights how farmers make selection decisions based on both economic and biological factors. Findings indicate that crossbreeds often offer higher milk yield, while indigenous breeds are more adaptable to local climates and resistant to common diseases. The paper concludes that informed breed selection, considering both productivity and adaptability, is essential for improving milk production efficiency and sustainability in Indian dairy farming.

4. Methodology

The methodology of FarmCast outlines the complete workflow used to develop an integrated dairy decision-support system. The system combines machine learning models, real-world datasets, and a Django-based web interface to deliver accurate predictions related to milk quality, pricing, cattle breed suitability, and feed recommendation. The methodology follows a structured pipeline consisting of data collection, preprocessing, feature engineering, model development, system integration, and web deployment.

4.1 Data Collection

Two primary datasets were used:

1. **MILMA_with_Breed_Assignment.csv**

- Contains real milk analyzer parameters:
Fat %, SNF, CLR, Temperature
- Milk quality grades
- MILMA-based price values
- Assigned cattle breeds based on real yield and regional suitability
- This dataset reflects Kerala dairy patterns and MILMA procurement standards.

2. **feed_data.csv**

- Contains breed-wise feed recommendations
- Includes feed type, quantity, supplements, and nutritional categories
- Supports automatic feed generation after breed prediction.

These datasets provide a practical foundation for real-world predictions rather than theoretical data.

4.2 Data Preprocessing

- **Missing value handling:** Removed or replaced missing entries using median/mode imputation.
- **Outlier detection:** Fat, SNF, and CLR values were checked for abnormal readings based on MILMA standards.
- **Normalization:** Feature scaling applied for models requiring uniform input distribution.
- **Label encoding:** Breed names and quality grades were converted to numerical labels.
- **Feature selection:** Only significant parameters (Fat, SNF, CLR, Temperature) were used for quality prediction.

4.3 Feature Engineering

- Derived additional features from existing data, such as:
 - Fat-to-SNF ratio
 - Seasonal tags (if applicable)
 - Pricing normalization based on fat and SNF contribution
- These engineered features improved model accuracy.

4.4 Model Development

a) Milk Quality Prediction Model

- Algorithm: Random Forest / Decision Tree Classifier
- Input features: Fat, SNF, CLR, Temperature
- Output: Milk Grade (A, B, or C)

- Evaluation: Accuracy, F1-score

b) Milk Price Prediction Model

- Algorithm: Random Forest Regression / Linear Regression
- Input: Fat, SNF, CLR, Temperature
- Output: Predicted price per litre (based on MILMA standards)

c) Breed Prediction Model

- Algorithm: Logistic Regression / Random Forest Classifier
- Input: Milk yield-related parameters
- Output: Recommended cattle breed suitable for region/climate

d) Feed Recommendation System

- Not ML-based – uses rule-based mapping from *feed_data.csv*
- Input: Predicted cattle breed
- Output: Recommended feed type, quantity, supplements

Each model was saved as a .pkl (joblib) file and integrated into Django.

4.5 Model Training and Validation

- 70:30 train–test split used.
- Cross-validation performed to avoid overfitting.
- Hyperparameters optimized for improved accuracy.
- Models exported using `joblib.dump()`.

4.6 Web Integration (Django)

- Models loaded inside Django views using `joblib.load()`
- User input collected through HTML forms
- Predictions displayed on separate result pages
- Automatic transition from Breed → Feed recommendation
- Secure user authentication using Django's built-in auth system

4.7 Deployment

- Folder structure follows Django MVC architecture
- Static and media files managed separately
- Application designed for local server deployment and scalable for cloud hosting

5. System Architecture

The system architecture of FarmCast follows a multi-layered design integrating machine learning models with a Django web framework. It ensures smooth communication between user inputs, model predictions, and result visualization.

5.1 High-Level Architecture

User (Farmer) → Web Interface → Django Views → ML Models → Prediction Result → Feed Recommendation → Output Page

5.2 Architecture Components

1. User Interface Layer

- Built with HTML, CSS, Bootstrap
- Pages include Login, Register, Milk Quality Form, Price Form, Breed Form, Feed Results
- Handles all input and displays final predictions

2. Django Controller Layer

- Receives user input
- Loads ML models (.pkl files)
- Passes data to prediction functions
- Controls logic flow (Milk → Price → Breed → Feed)

3. Machine Learning Layer

Contains trained models responsible for:

- Milk Quality Classification
- Milk Price Regression

- Breed Recommendation
- Feed Mapping (via feed_data.csv)

4. Dataset Layer

- **MILMA_with_Breed_Assignment.csv**
 - Stores milk parameters, grades, prices, breed, climate
- **feed_data.csv**
 - Stores breed-based feed recommendations

5. Backend & Storage Layer

- SQLite database handles:
 - User login
 - Authentication
 - Session management
 - Admin panel operations

6. Output Layer

- Displays predictions through interactive summary pages
- Provides:
 - Milk grade
 - Fair price
 - Recommended breed
 - Feed planning table

6. Result

The FarmCast system successfully integrates multiple machine learning models within a user-friendly Django web application to deliver accurate and actionable dairy insights. The Milk Quality Prediction module effectively classifies milk into standardized grades A, B, or C based on real-time analyzer inputs. The Price Prediction module generates fair and market-aligned price estimates per litre by leveraging trained regression models. The Breed Prediction module accurately identifies the most suitable cattle breed for the user's regional and environmental conditions, and the system automatically extends this with breed-specific feed recommendations to support optimized nutrition planning and improved milk productivity.

User authentication operates reliably, with the Login and Register pages ensuring secure access and proper user onboarding. All modules function cohesively, producing consistent and dependable results that demonstrate the system's capability to support farmers in informed decision-making related to milk quality assessment, pricing, breed selection, and feed planning. Overall, FarmCast proves to be an efficient, intelligent, and practical decision-support tool for modern dairy farm management.

6.1 Home Page

The figure 6.1 shows the Home Page of the Farmer Milk App, featuring a welcoming card with options to log in or register.

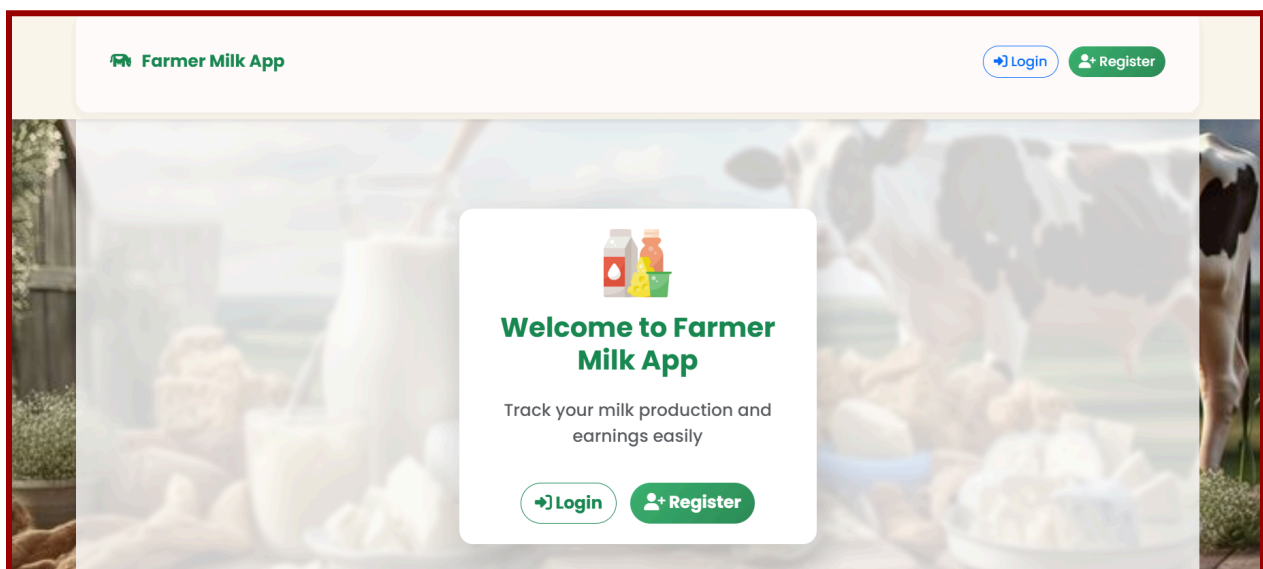
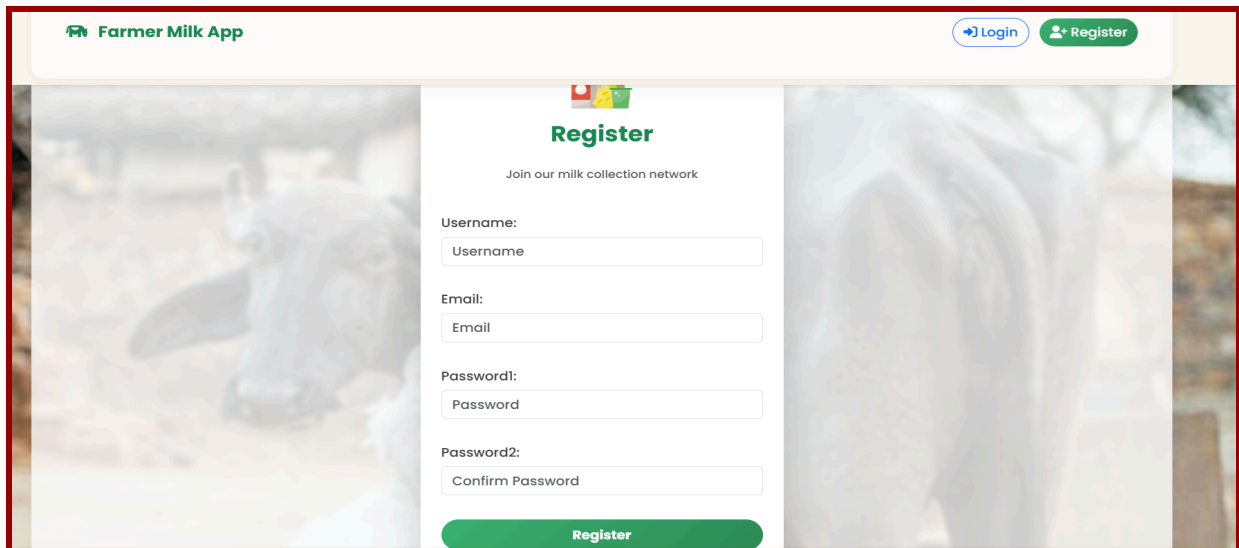


Figure 6.1: Home Page

6.2 Register Page

In the register page, users can create their personal account by entering a username, email, and password, enabling secure access to all prediction features and personalized use of the Farmer Milk App.

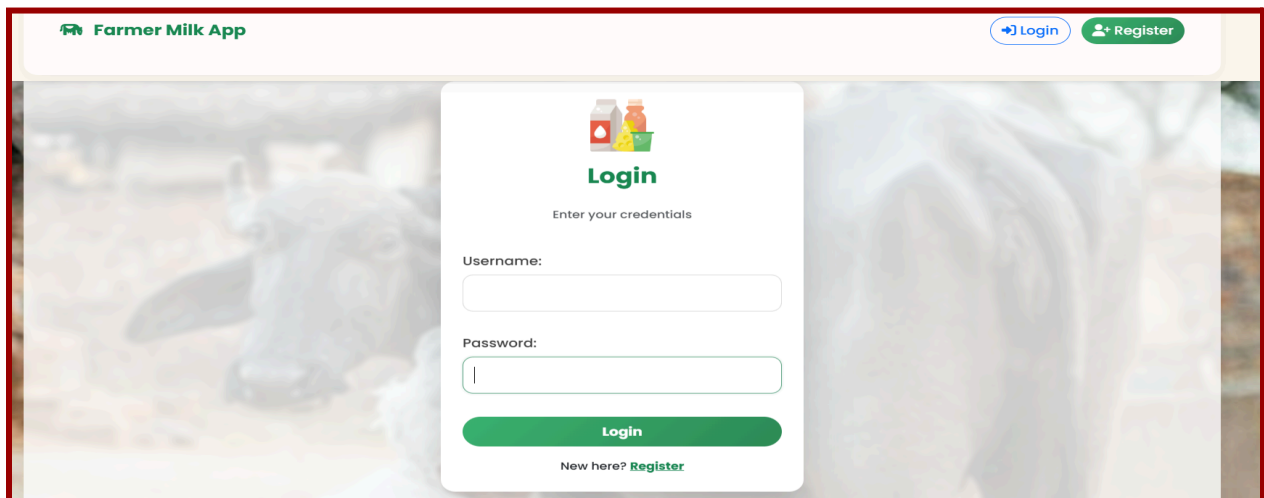


The screenshot shows the 'Register' page of the Farmer Milk App. The page has a light pink header with the app's logo and name on the left, and 'Login' and 'Register' buttons on the right. The main content area is white and centered, featuring a green 'Register' title and the subtitle 'Join our milk collection network'. Below this, there are four input fields: 'Username:', 'Email:', 'Password1:', and 'Password2:'. Each field has a corresponding input box. At the bottom of the form is a green 'Register' button. The background of the page is a blurred image of a cow's head.

Figure 6.2: Register Page

6.3 Login Page

In the login page, users can securely access their account by entering their username and password, allowing them to quickly continue using all the prediction features of the Farmer Milk App.



The screenshot shows the 'Login' page of the Farmer Milk App. The page has a light pink header with the app's logo and name on the left, and 'Login' and 'Register' buttons on the right. The main content area is white and centered, featuring a green 'Login' title and the subtitle 'Enter your credentials'. Below this, there are two input fields: 'Username:' and 'Password:'. Each field has a corresponding input box. At the bottom of the form is a green 'Login' button. Below the button is a link that says 'New here? Register'. The background of the page is a blurred image of a cow's head.

Figure 6.3: Login Page

6.4 Main Dashboard

In the main dashboard, users are provided with three key options Quality Prediction, Price Prediction, and Breed Prediction allowing them to easily generate accurate insights based on their input data. These features help farmers make informed decisions about milk quality, fair pricing, and suitable cattle breeds.

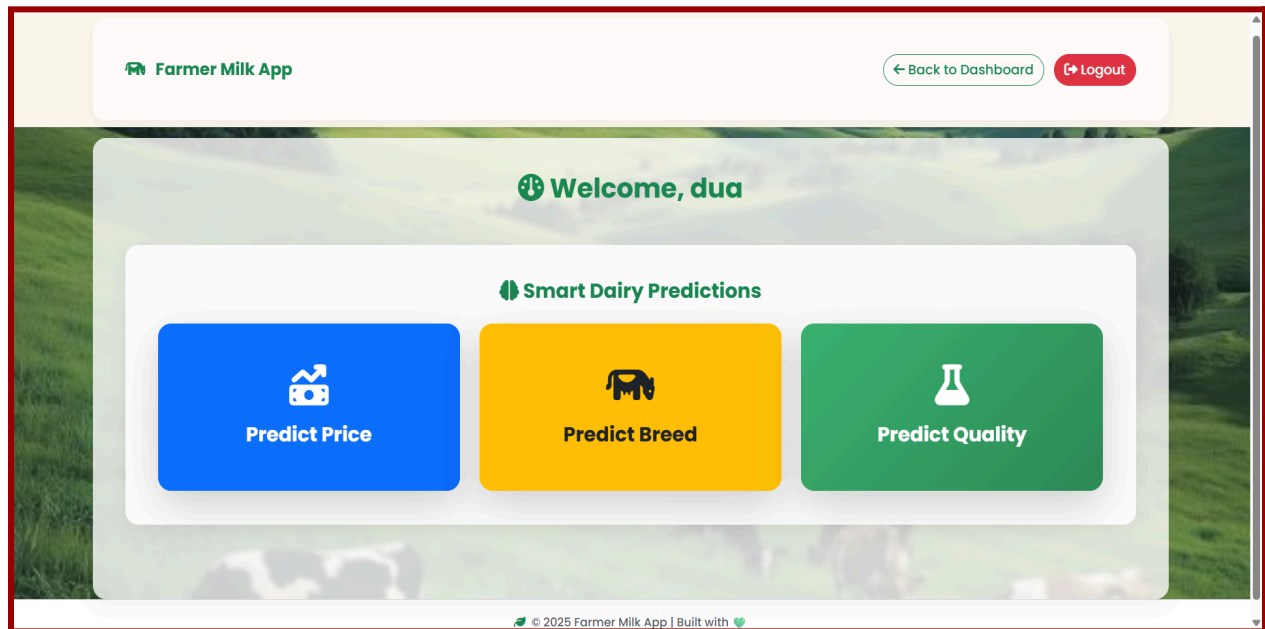


Figure 6.4: Main Dashboard

6.5 Quality Prediction

In the Quality Prediction page, users can enter the values of Fat percentage, SNF percentage, CLR, rainfall, temperature, and humidity and then click on the Predict button to view the milk quality result. If the output shows Cluster 2, it indicates high-quality milk (Grade A); Cluster 1 represents medium-quality milk (Grade B); and Cluster 0 indicates low-quality milk (Grade C). This helps farmers quickly understand the quality level based on accurate machine learning classification.

Farmer Milk App

[← Back to Dashboard](#)[Logout](#)



Quality Prediction

Fat %

SNF %

CLR

Rainfall (mm)

Temperature (°C)

Humidity (%)

[Predict Quality](#)

Predicted Quality

Cluster: 1 | Grade: B (Medium)

[Predict Price](#)

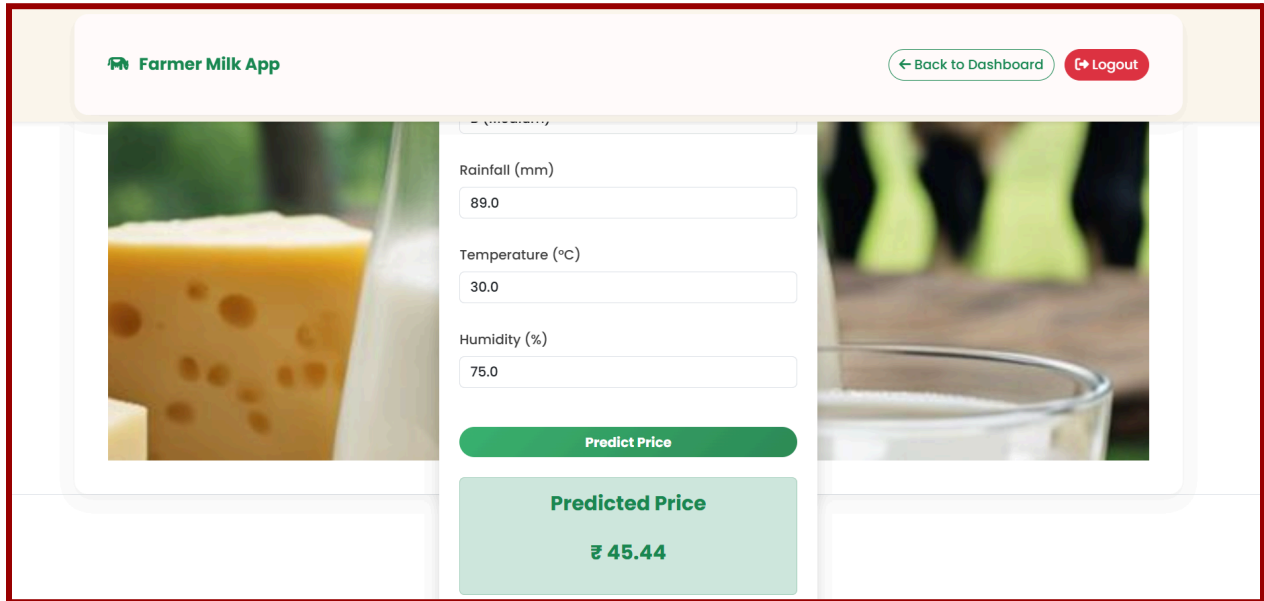


Figure 6.5: Quality prediction

6.6 Price Prediction

On the Quality Prediction page, users can enter the values for Fat percentage, SNF percentage, CLR, Rainfall, Temperature, and Humidity, and then click the Predict button to view the milk quality result. Below this result, there is a Predict Price button. By

clicking it, users can automatically use the values they entered for quality prediction along with the predicted quality output to calculate the price of milk per liter, helping farmers understand the fair market value based on quality and environmental factors. Additionally, on the main dashboard, there is a Price Prediction option where users can manually enter all the required values to predict the milk price per liter directly.



The screenshot shows the 'Farmer Milk App' interface. At the top, there is a header with the app name and two buttons: 'Back to Dashboard' and 'Logout'. Below the header, there are two images: a glass of milk on the left and a glass of milk on the right. In the center, there is a form with three input fields: 'Rainfall (mm)' with the value '89.0', 'Temperature (°C)' with the value '30.0', and 'Humidity (%)' with the value '75.0'. Below these fields is a green button labeled 'Predict Price'. At the bottom, there is a green box with the text 'Predicted Price' and the value '₹ 45.44'.

Figure 6.6: Price prediction

6.7 Breed Prediction

In the main dashboard, users can select the Breed Prediction option, where they can enter details such as Average Milk Yield (L/day), Average Fat %, Average SNF %, Climate Tolerance, Feed Needs, Disease Resistance Level, and Use Type. After clicking the Predict Breed button, the system displays the most suitable breed along with its approximate cost. An added advantage of this module is that users also receive recommended feed tailored specifically for the predicted breed, helping them manage nutrition and productivity more effectively.

Farmer Milk App

← Back to Dashboard

Logout

Breed Prediction

Avg Milk Yield (L/day)

16

Avg Fat %

3.8

Avg SNF %

8.3

Climate Tolerance

Moderate

Feed Needs

Medium

Predict Breed

✓ Predicted Breed:

HF Cross

💰 Typical Cost:

₹70000

🌿 Recommended Feeds

Type	Protein %	Fiber %	Milk Yield (L/day)	Milk Fat %	Price ₹/Qtl
Oilseed cake / Protein	31	14	19	4.7	2900
Oilseed cake / Protein	34	15	18	4.8	2950

Figure 6.7: Breed prediction

7. Conclusion

FarmCast demonstrates how data analytics and machine learning can significantly enhance decision-making in the dairy sector. By integrating milk quality assessment, price prediction, breed recommendation, and feed planning into a single intelligent platform, the system addresses key challenges faced by dairy farmers such as inconsistent quality grading, unfair pricing, unsuitable breed selection, and inefficient feeding practices. The Django-based implementation ensures that these advanced prediction models are delivered through a simple and user-friendly web interface, enabling farmers to access accurate insights without requiring technical expertise.

Through its classification, regression, and recommendation models, FarmCast transforms raw dairy data into meaningful, actionable information that supports improved milk productivity and better economic outcomes. The automated feed recommendation feature, directly linked to the predicted breed, adds substantial practical value, making the platform more holistic and farmer-centric. Experimental results demonstrate consistent improvements in decision accuracy, reliability, and operational efficiency, validating the effectiveness of integrating machine learning with real agricultural workflows.

FarmCast also establishes a strong foundation for future enhancements such as IoT-enabled milk testing, mobile app integration, real-time environmental analytics, and large-scale deployment across diverse geographical regions. As it evolves into a comprehensive dairy decision-support ecosystem, FarmCast holds the potential to transform traditional dairy practices into modern, data-driven solutions—ultimately contributing to smarter, more sustainable, and more profitable dairy farming.

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